

## Assignment:

This assignment will contribute 30% to your overall mark for ELEC345.

### This is a team assignment as follows:

1. Form into teams of 2-3 students (maximum) and select a team leader.
2. You are to complete several RF design tasks and create supporting documentation to go with them. Firstly, you will each create a dipole antenna. Then as a group design a more complex antenna using copper wire on a structure or copper tape on a choice of backing. You will need to test and create a data sheet for it. You will also need to design and do various tests on an RF module to industry standards and compare to the data sheet to ensure it is operating as you expect and on the same board a chip antenna.
3. For your designs/modules, produce a technical note and data sheet that will introduce colleagues to the necessary theory (with references) and guide them through the work. You should enhance this with test results and simulations (proteus/ansoft designer/Altera /Matlab/Mathcad etc) wherever possible.
4. **Submit a copy by 0900 Monday 8<sup>th</sup> December the day of the presentations and** Include in the final copy a single page summary describing:
  - contributions from team members and how the team worked together - the presentation plan
5. Present your design guide to colleagues during the session week starting. **9th December.** Twenty-minute time slots followed by 5-10 minutes of questions will be allocated.

### Assessment will be as follows:

| Element                     | Weight | Criteria                                                                                                |
|-----------------------------|--------|---------------------------------------------------------------------------------------------------------|
| Technical note & data sheet | 50%    | Work completed, detail and testing.<br>Theory, results explained.<br>Accurate & referenced.<br>Concise. |

|                         |        |                                                                                                                                                                   |
|-------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presentation            | 50%    | Interesting and engaging<br>Good visual quality Good<br>voice style, practiced.<br>Accurate & correct.<br>To time and well structured<br>Able to answer questions |
| Individual Contribution | Factor | Report recommendation (staff moderated)                                                                                                                           |

List of tasks:

1. Build a half wavelength dipole and test it.
2. Design and build your main antenna for the report and data sheet.
  - i Measure and test it for the data sheet values.
  - ii Measure its radiation pattern.
  - iii Transmit down the length of the lab and look at the received power and comment on the link budget.
3. Design and test the pcb.
  - a. Measure chip antenna impedance (S11, smith chart & return loss)
  - b. Calculate values for a matching network, implement it and test it.
  - c. Measure S11 of 20dB gain stage (return loss) and gain S21.
  - d. Measure the 1dB compression point.

Deliverables:

2. Technical note. (Antenna and PCB)
  - a. Design
  - b. Testing & results
  - c. Explanation & theory.
3. Data Sheet. (Antenna)
  - a. Delete the values in one of the older chip antenna data sheets, insert your own as much as you can.
4. Presentation
  - a. Summarize you design and results and explain them.